

VEDANT LALIT BHATT

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

B.S. in Computer Science | GPA: 3.8 — Expected Graduation: May 2027

August 2024 – Present

- **Concentrations:** Intelligence and Information Networks
- **Relevant Coursework:** AI, Computer Systems and Networks, Data Structures and Algorithms, Database Systems, OS
- **Scholarships & Awards:** Zell Miller Scholarship, Georgia Certificate of Merit, IT Specialist Certiport Software Development

EXPERIENCE

BrainLab Research Group

Atlanta, GA

Undergraduate ML Researcher

September 2025 – Present

- Built Python pipelines to preprocess EEG signals using filtering, epoching, and artifact removal for brain-computer interfaces
- Engineered time series models to classify brain waves and SSVEP activity, achieving 89% accuracy via cross-validation
- Applied adaptive noise cancellation to remove power line interference, improving signal fidelity via 28% noise reduction

Quilly

Atlanta, GA

Lead Software Engineer Intern

August 2025 – Present

- Led team of 7 developer interns, coordinating daily stand-ups and sprint plannings to streamline collaborative development
- Optimized PostgreSQL with indexing, batch writes, and connection pooling, improving read times by 75% and writes 60%
- Increased platform performance by improving Redis cache hit rate by 40%, reducing data fetch latency and server load

King of the Curve

Remote

Software Engineer Intern

May 2025 – August 2025

- Refactored 20+ Flutter components using Riverpod for state management, iterating features collaboratively with Agile sprints
- Optimized cross-platform app load times by 30% through image compression, lazy loading, and asynchronous data fetching
- Implemented offline data caching, reducing app crashes by 40% for 4,000+ monthly users in low-connectivity areas

Big Data Big Impact - FishCast

Atlanta, GA

Data Engineer, Project Manager

February 2025 – Present

- Built ETL pipelines processing NOAA and Fishbase data using Python, enabling predictive modeling of marine movement
- Trained LSTM models on habitat data, achieving 87% accuracy with 5-fold cross-validation to forecast Atlantic Cod shifts
- Designed GIS dashboards on AWS EC2 using Lambda for preprocessing, applying event-driven architecture for insights

Alpha Kappa Psi (Professional Fraternity)

Atlanta, GA

Director of Software Development

January 2025 – Present

- Led team of 8 developers to create a React website used by 200+ members, streamlining member tracking and performance
- Implemented WebSockets for real-time work, integrating MongoDB for message storage and driving latency under 100ms
- Refactored inefficient REST routes by minimizing overfetching, indexing high-traffic queries, accelerating performance 2.2x

PROJECTS

EEG Emotion Classification — Python | TensorFlow | NumPy | SciPy | MATLAB | Git

October 2025

- Developed a pipeline extracting delta, theta, alpha, beta, and gamma band features from multichannel EEG signals using FFT
- Trained a shallow neural network on normalized band power features, achieving 81% accuracy in emotion classification
- Applied artifact rejection for noise and two-second epoch segmentation for cleaned signals, improving model analysis

FileAI — TypeScript | Python | OpenAI CLIP | HuggingFace | FAISS | CedarOS | NumPy | GraphQL | Git

September 2025

- Built an interactive natural language chatbot enabling users to execute file actions: create, move, organize, zip, and search
- Implemented semantic file parsing, mapping file contents to vector embeddings for efficient representation and retrieval
- Accelerated query times 75% by caching and retrieving previously calculated vector embeddings with FAISS Library

MySQL - Runner — TypeScript | Node.js | Python | Flask API | MySQL | Ollama | Git

August 2025

- Developed a VS Code extension integrating Ollama's Llama3.2 NLP model to convert natural language queries into SQL
- Increased output accuracy by 30% using schema-aware prompts to ensure valid table and column names in generated queries
- Built to boost productivity by seamlessly integrating multiple tools, reducing context switching and saving navigation time

SKILLS

Languages: Java | HTML | CSS | Python | Assembly | Swift | SwiftUI | C# | C | C++ | SQL | JavaScript | TypeScript

Software: Xcode | Git | Excel | Word | Powerpoint | Jira | Axios | Visual Studio Code | PyCharm | IntelliJ IDEA | Terminal/Bash

Frameworks/Platforms: React.js | Hugging Face | Linux | Pandas | Flask | REST | AWS | GraphQL | Docker | Node.js | RDBMS